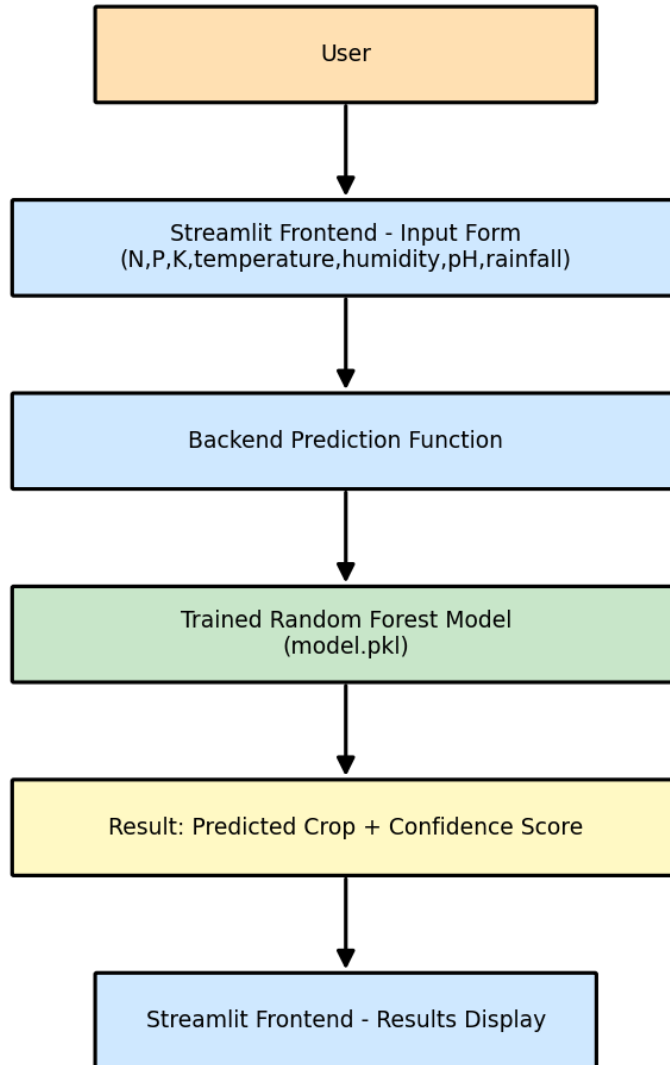

Requirement Analysis

Data Flow Diagram & User Stories

Date	28 June 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Data Flow Diagram:

OptiCrop - Data Flow Diagram



User Stories:

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release

Farmer	Input Collection	USN-1	As a farmer, I can enter my soil's N, P, K values and local weather conditions into the form	I can fill all 7 fields and click the recommended button	High	Sprint-1
Farmer	Crop Prediction	USN-2	As a farmer, I want to see the recommended crop with a confidence score	I see a clear crop name and confidence percentage after clicking the button	High	Sprint-1
Farmer	Alternate Suggestions	USN-3	As a farmer, I want to see top-3 alternate crop options in case the first one doesn't suit me	I see a ranked list of 3 alternate crops	Medium	Sprint-2
Agricultural Officer	Bulk Testing	USN-4	As an officer, I can test multiple different soil samples one after another without reloading the page	I can clear fields and enter new values immediately	Medium	Sprint-2

User	Error Handling	USN-5	As a user, I get a clear error message if I enter a wrong or out-of-range value	I see a readable error message and the app does not crash	High	Sprint-1
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